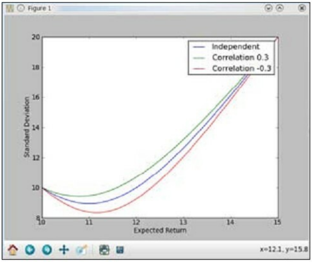

Figure 1: A graph of the standard deviation as a function of the return, based on a basket of two investments. The graphs for three values of correlation between the two investments are shown here.

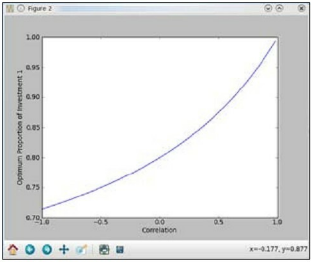

Figure 2: The graph shows how the optimum proportion for an investment varies as the correlation between two investments varies.

```
plt.plot(avg_return(p),std_dev(p, 0),'-', label='Independent')
plt.plot(avg_return(p),std_dev(p, 0.3),'-', label='Correlation 0.3')
plt.plot(avg_return(p),std_dev(p, -0.3),'-', label='Correlation -0.3')
plt.xlabel('Expected Return')
plt.ylabel('Standard Deviation')
plt.legend()
plt.show()
```

The safest investment has a mixture of both and gives a better return than the safer of the two.

The best proportion to use will be the one that minimises the standard deviation. It is easy to get that value. Just add the following code:

```
from scipy import optimize
res = optimize.fmin(std_dev, 50, args=(0,))
print "Result", res
```

The *fmin* function in the *optimize* module accepts a function, *std_dev* in your case, as a parameter and a starting guess. Additional parameters for the *std_dev* function are passed using the *args* parameter. It then finds the optimal value. This ought to give you the result that you should have about 80 per cent in the first investment and 20 per cent in the second as the safest option, in case the investments are independent:

```
Optimization terminated successfully.
         Current function value: 8.944272
         Iterations: 23
         Function evaluations: 46
Result [ 0.80001831]
```

You know that the investments are usually correlated. So, you may want to examine, visually, how the result varies depending upon the correlation. Add the following code to compute the optimum proportion of Investment 1 for a list of correlations and plot it:

```
def optimum_p(corr):
    res = []
    starting_val = 50
    for c12 in corr:
        res.append(optimize.fmin(std_dev,starting_val,args=(c12,)))
        # use the previous result as a starting guess
        starting_val = res[-1]
    return res
corr_values = np.arange(-1, 1., 0.02)
res = optimum_p(corr_values)
fig = plt.figure()
fig.canvas.set_window_title('Figure 2')
plt.plot(corr_values,optimum_p(corr_values),'-')
plt.xlabel('Correlation')
plt.ylabel('Optimum Proportion of Investment 1')
plt.show()
```

The result is shown in Figure 2. If both investments are perfectly correlated, then you are better on all counts with Investment 1. However, if they have a perfect negative correlation, the best option is to invest only 70 per cent in Investment 1 and the remaining in Investment 2. An example of negative correlation is the frequently heard advice: "When stock markets are booming, interest rates fall."

If only finding the correct model in economics was easier!

LP—finding a solution

Let's suppose that the government can spend Rs 100 on increasing the consumption and production of food. It estimates that each rupee spent on rural employment will increase consumption by a rupee. This increase must be matched by an increase in production. It estimates